

Image Edge Detection Using Fuzzy Logic

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Abstract—This document presents a fuzzy logic based edge detection algorithm using first-order derivation of pixel values. Edge detection is an important step in image processing with uses ranging from medical imaging to areal surveillance. The proposed method is an edge detection algorithm containing a fuzzy logic system that produces a clear image of the edges present in the input image. The project is implemented in MATLAB and experimental results will be presented. Comparisons between the proposed solution and other common edge detection methods including Sobel, Prewitt, Roberts, Log, Zerocross and Canny based methods will be explored. Further comparisons between the proposed method and results obtained by similar papers will be achieved through visual inspection as well as entropy calculations. Finally an explanation of the difficulties faced and work that could be done for future improvements of the project will be explored.

Index Terms—Edge detection, Entropy, Fuzzy logic, Image processing

I. INTRODUCTION

Object detection and recognition is an important component in the field of image processing because of its wide use in various professional fields. From finding cancer in medical imaging to helping inspectors locate cracks in a buildings foundation, object detection is a vital component of modern engineering. The first step in recognising objects in an image often involves edge detection. Edge detection is an image processing step that locates high frequency changes in pixel intensity. These high frequency changes are called edges because in reality, boundaries between objects usually have different luminosities. This difference in luminosity will show up in the captured image and are able to be extracted as edges. These edges could be sent for additional processing to a machine learning

algorithm that can classify the objects present in the image.

There are many algorithms that have been proven to be effective in a wide range of scenarios. MATLAB provides their own implementation of these algorithms for use in their software. They include Sobel, Prewitt, Roberts, Log, Zerocross and Canny based algorithms. By default, MATLAB thins the output image when using these algorithms. Thinning is done to produce a cleaner result with less noise. In addition to these common algorithms provided in MATLAB, Fuzzy Logic has proven its ability to accurately detect edges in an image, providing similar results to the commonly used methods. For the purposes of this project, edge detection was performed in grayscale to simplify the implementation. This means that some edge data will be lost for boundaries with the same grayscale colour representation on either side.

This project will be based on the fuzzy logic based edge detection algorithm implementation presented in [1]. Changes will be made in the attempt to improve the algorithm with testing and analysis done to determine if the improvements were successful.

A. Fuzzy Logic

Fuzzy logic is a method of controlling outputs given an input which can be categorized by its

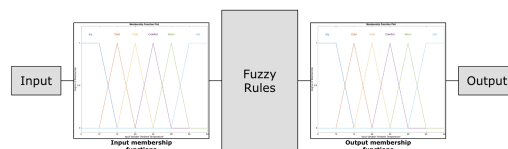


Fig. 1: Fuzzy logic system overview.

membership to various input parameters. This categorization membership is normalized to a range of 0 to 1 and allows for an input variable to be part of multiple memberships at the same time. Doing so allows for a more intuitive approach to designing image processing systems by allowing for text based rules to be represented by these membership functions and inputs. These text based rules allow for fast implementation and processing by using limited if and else statements. Fuzzy based systems are ideal for this application of image edge detection because the varying intensity each pixel has can be represented more accurately with a varying degree membership to a variable rather than a binary based membership.

Fuzzy systems have three main components which are fuzzification, fuzzy inference and defuzzification. These components can be seen in Fig. 1 where the input and output range is divided into various membership functions. The fuzzification system transforms inputs into various fuzzy sets called membership functions. The input can belong to more than one of these sets and have a varying degree of membership to those functions. The fuzzy inference system uses rules given by the programmer to map input member functions to outputs. These rules need to be carefully chosen to correctly model the desired behaviour the system is trying to achieve. The last step in a fuzzy logic controller is defuzzification. This step creates a crisp output by combining the degree of association to each output membership function. There are different methods used in the defuzzification step to obtain crisp output values. This project will utilize centroid defuzzification which computes the "center of gravity" after a max aggregation of all the fuzzified outputs obtained from the rule base. The proposed system will be a Mamdani Type-1 fuzzy inference system due to its intuitive rule inference [2].

B. Problem Definition

Utilizing knowledge of fuzzy logic systems an edge detection algorithm will be developed. This will require the detection of high frequency pixel changes which suggests that it might be useful to calculate the gradient of the image. The gradient represents the first order rate of change in pixel

values. If the algorithm is able to be developed successfully, comparisons will need to be done to compare the effectiveness of the proposed solution against the commonly used algorithms.

C. Objectives

The project outlined in this paper has the following objectives:

- 1) Research into traditional and fuzzy logic based edge detection algorithms
- 2) Implement a fuzzy logic system to detect edges in an image
- 3) Determine the effectiveness of the proposed solution through comparison techniques found in [3] (visual and entropy)
- 4) Compare the proposed solution to the solution presented in [1]

II. LITERATURE SURVEY

A literature survey was conducted to learn more about edge detection and fuzzy logic systems. [3] gives insight into the fundamentals of fuzzy logic systems and presents a fuzzy logic based edge detection algorithm. This project is based on the implementation in [1] with modifications in the attempt of exceeding its performance. The author of this paper presents an edge detection algorithm using a fuzzy logic system and calculating six gradients in the image. It also outlines a method of comparing the results of various edge detection algorithms by calculating the entropy of an image. [4] and [5] present fuzzy logic edge detection algorithms by calculating the radial gradient of each pixel. [6] explores the relationship between different measurement parameters of various common edge detection algorithms. [7] describes the thinning process its use in image processing. Further insight into entropy calculations and its use in determining the randomness of an image is presented in [8]. Finally, [9] and [2] were used as a guide to create the fuzzy logic system as they provide insight into MATLAB implementation of fuzzy logic systems. Through the completion of this literature survey, knowledge was gained into the topics needed to develop fuzzy logic systems as well as compare the effectiveness of various image processing algorithms.

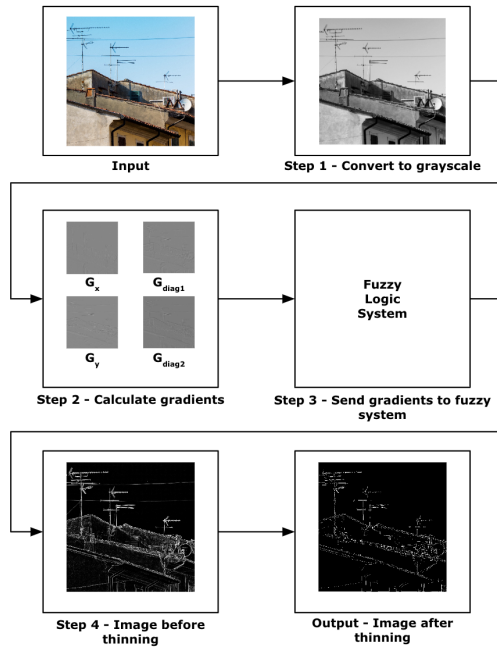


Fig. 2: Implemented solution flow diagram.

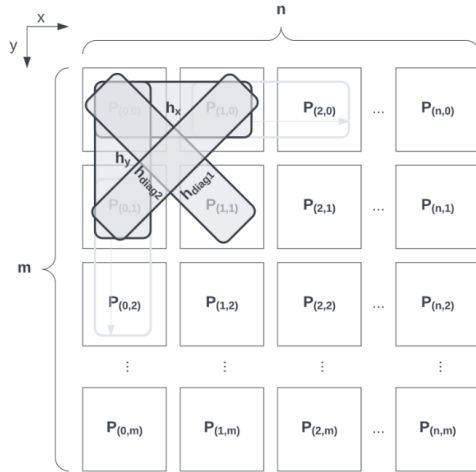


Fig. 3: Image convolution with moving filters.

III. IMPLEMENTED SOLUTION

A. Edge Detection Algorithm

An edge detection program was created in MATLAB and a flow diagram of the implemented solution can be seen in Fig. 2. In order to perform edge detection on an image it is first converted into grayscale. This is done to simplify the algorithm while still being able to detect edges in the image. For the purpose of this project it is assumed that the edges in the image can be seen when converting

the image to grayscale. This means that there is contrast on either side of the edge. Edges can be detected by locating high frequency changes in pixel values in the image. As proposed in [1], calculating the gradient of the image in the horizontal and vertical directions will reveal these high frequency regions. As shown in Fig. 3, the gradient in the x (G_x), y (G_y), diag1 (G_{diag1}) and diag2 (G_{diag2}) are calculated by performing a convolution with h_x , h_y , h_{diag1} and h_{diag2} on all pixels ($P_{(n,m)}$) in the image. The convolution uses the moving filter h_x , h_y , h_{diag1} or h_{diag2} to subtract the current pixel from its neighbour. The h_x filter moves horizontally across the image, calculating the gradient for each row. The h_y filter performs the same operation but in the vertical direction, calculating the gradient for each column. The h_{diag1} and h_{diag2} filters perform the same operation but in the two diagonal directions. All of these filters are defined in eq. (1) to (4) and are used for the convolutions in the x, y and two diagonal directions. After these gradients are found, the data is fed into the fuzzy logic system.

$$h_x = \begin{bmatrix} 1 & -1 \end{bmatrix} \quad (1)$$

$$h_y = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad (2)$$

$$h_{diag1} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (3)$$

$$h_{diag2} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad (4)$$

B. Fuzzy Logic System

The designed fuzzy logic system contains four inputs corresponding to the gradients calculated previously. As seen in Fig. 4a to 4d, the inputs each contain one Gaussian membership function centered at zero to represent the inputs degree of membership to being zero. This function is given by eq. (5) with $c = 0.2$ which represents the standard deviation. The value of c was chosen experimentally with guidance from [2] and controls the width of the Gaussian function, thus altering the sensitivity of the system to non-zero gradient inputs. A larger value of c will make the system less sensitive to changes in pixel intensity, decreasing the intensity of detected edges [2].

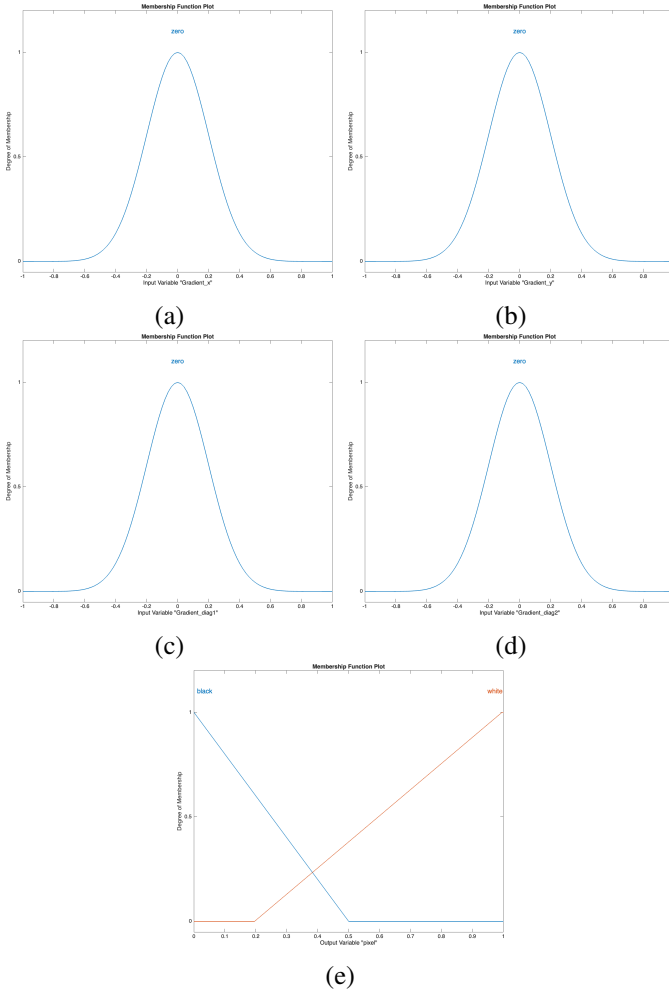


Fig. 4: Input/output membership functions, (a) x-gradient, (b) y-gradient, (c) diag1-gradient, (d) diag2-gradient, (e) pixel.

$$f(x) = e^{-\left(\frac{x^2}{2c^2}\right)} \quad (5)$$

The system also contains one output corresponding to the pixel intensity at the current location. As seen in Fig. 4e, the output contains two triangular membership functions corresponding to a black or white pixel where a white pixel represents an edge.

There are two rules which map the fuzzy inputs to fuzzy outputs. These rules state that if all gradients are zero, the pixel is black, meaning there is no edge. If any gradient is not zero, the pixel is white and there is an edge. The rules are defined as follows:

- 1) If G_x is zero and G_y is zero and G_{diag1} is zero and G_{diag2} is zero then pixel is black

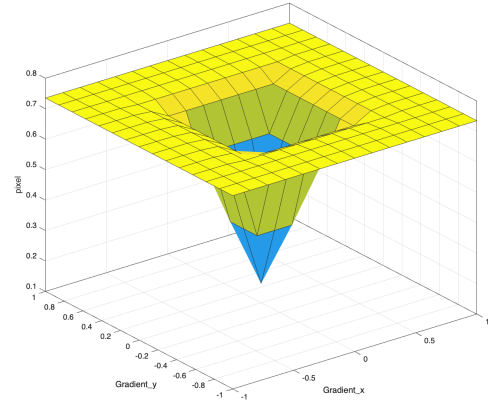


Fig. 5: Control surface for x-gradient vs. y-gradient vs. pixel.

- 2) If G_x is not zero or G_y is not zero or G_{diag1} is not zero or G_{diag2} is not zero then pixel is white

A control surface is obtained by plotting all possible inputs and outputs on a graph. Control surfaces are useful to confirm that the designed fuzzy logic system is performing as expected by allowing for a visual representation of the system. As seen in Fig. 5, the crisp pixel output value is higher when the gradients are not zero and represents an edge in the output image. The control surfaces for each gradient input combination are identical due to the rule set presented earlier, so only one combination is shown here.

C. Thinning Morphological Operator

The final step in the algorithm presented in this project is “thinning”. This operation reduces the stroke width of the lines in the output image. This process reduces unnecessary data in the image and produces a cleaner result. Image thinning has been used in various fields including medical, construction and fingerprint analysis. The differences between the output image before and after thinning can be seen in Fig. 6. From this figure it can be seen that some edge data is lost during the thinning operation. This is most likely due to discontinuities in the edge pixels causing the algorithm to remove pixels that should be kept. These discontinuities are most likely caused by the sensitivity of the fuzzy logic system and the quality of the image being analysed. It can be seen that the thinning operator does better with the roof image which can

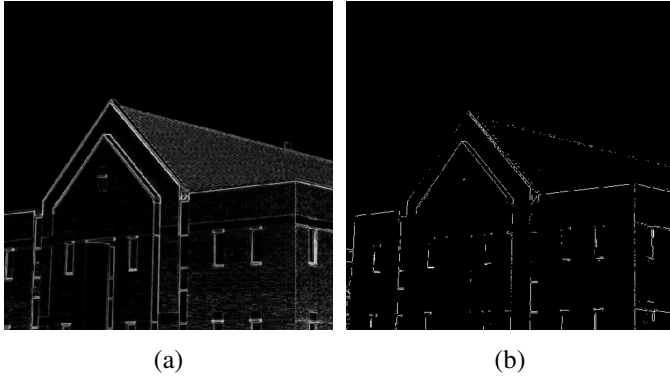


Fig. 6: Output image thinning comparison, (a) before thinning , (b) after thinning.

be attributed to the higher pixel count that this image has which allows for a more pronounced gradient.

IV. RESULTS

A. Horizontal and Vertical Gradients

The first result uses the input image of the roof seen in Fig. 2 with only the horizontal and vertical calculated gradients. The output can be seen in Fig. 7a. These gradients are sent to the fuzzy logic controller which has a simplified rule base to only utilize these two gradient inputs in the calculation of the output pixel value. Visual inspection of the results show that the proposed method produces a comparable result to the commonly used edge detection methods presented earlier. The prominent edges of the image are clearly visible.

B. Horizontal, Vertical and Diagonal Gradients

The following result also uses the input image of the roof seen in Fig. 2 with the horizontal, vertical and two diagonal gradients. The output can be seen in Fig. 7b. These four inputs are sent to the fuzzy logic controller which has the full rule base specified in Section III-B. Visual inspection of the proposed result compared to the commonly used algorithms shows that a similar result is obtained as seen in Section IV-A. When viewing the image closely, it can be seen that there is more accurate edge data for edges that are not at 0° or 90° . This is evident in the image where the antenna edges that are near 45° are more pronounced than in the previous solution. This implementation shows that including calculations of the diagonal gradients

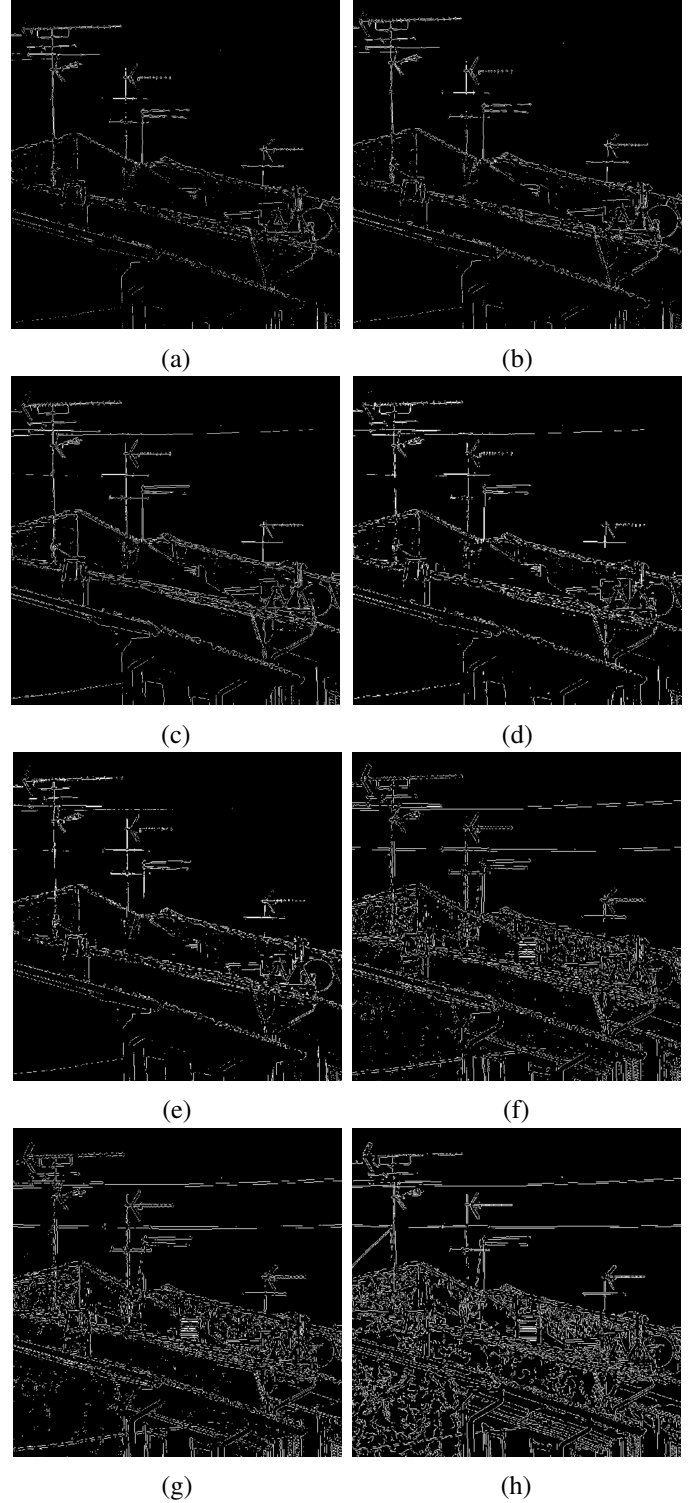


Fig. 7: Output image with different algorithms, (a) proposed method using G_x and G_y , (b) proposed method using G_x , G_y , G_{diag1} and G_{diag2} , (c) sobel, (d) prewitt, (e) roberts, (f) log, (g) zerocross, (h) canny.

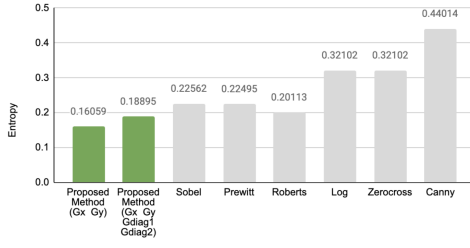


Fig. 8: Entropy comparison of proposed solutions with roof image.

allow the fuzzy logic controller to perform better by producing a more desirable result.

C. Entropy Comparison

Calculating the entropy of an image can be used in conjunction with visual inspection to determine how well an edge detection algorithm is performing by giving a statistical value for the randomness of the image [8]. A lower number suggests that the image is less random. In addition to image data which contributes to the clarity of the scene, images will contain some amount of unwanted random noise. In general, pixel randomness in an image is a good determination of how much unwanted data is present. Results in Fig. 8 show calculated entropy values for the proposed methods as well as the commonly used edge detection algorithms. From this figure it can be seen that the proposed solution has comparable entropy values to the commonly used algorithms. This figure also shows that the proposed solution produces a result similar to Sobel, Prewitt and Roberts based algorithms. Data from Log, Zerocross and Canny based algorithms have a noticeably higher entropy value which suggests they contain more unwanted data. These entropy calculations confirm the results seen through visual inspection where the Log, Zerocross and Canny based implementations produce a result with edge data that should be discarded as noise.

D. Comparison to Paper

Comparing the results obtained in this project with results from [1], similar results can be seen through visual inspection as seen in Fig. 9. In this section, the same image of a house was processed with the intention of accurate comparisons between the proposed algorithm and the algorithm presented

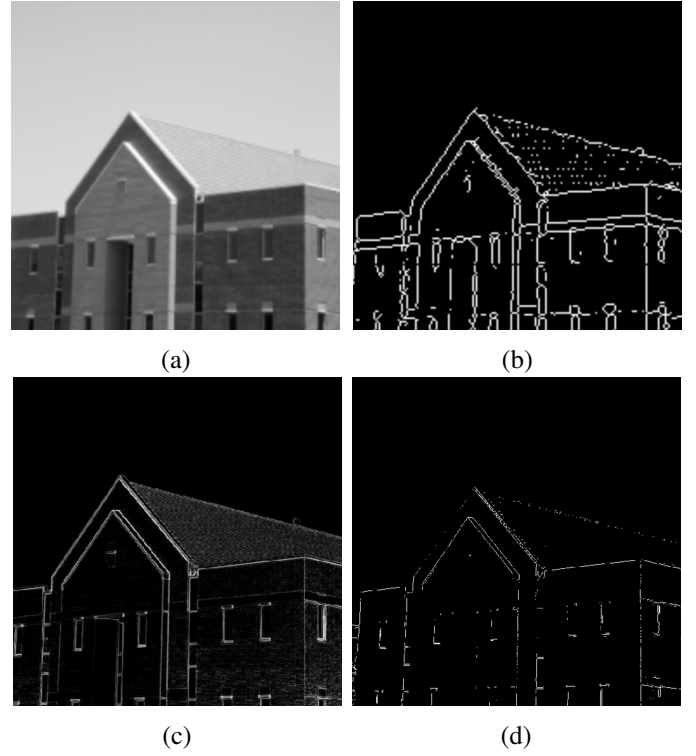


Fig. 9: Results with house image, (a) input image, (b) output image from [1], (c) proposed method using G_x , G_y , G_{diag1} and G_{diag2} before thinning, (d) proposed method using G_x , G_y , G_{diag1} and G_{diag2} after thinning.

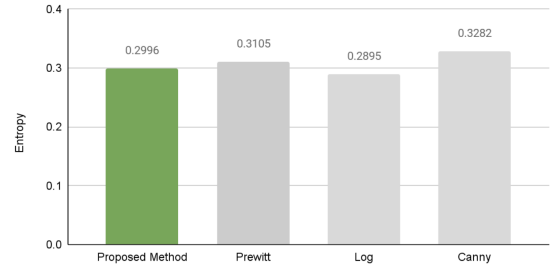


Fig. 10: Entropy for different edge detection methods from [1] using house image.

in [1]. It can be seen through visual inspection that the proposed solution is able to accurately extract the edge data in the image. The results before and after thinning show that the algorithm reduces unnecessary edge data and produces a cleaner result. Entropy calculations from [1] are seen in Fig. 10 and show that their implementation produces similar results to other commonly used algorithms. Entropy calculations using the proposed solution with the

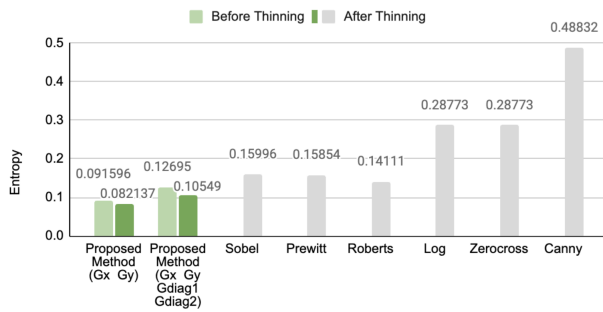


Fig. 11: Entropy for various edge detection methods using proposed method on house image.

house image are seen in Fig. 11. These results show similar entropy values for the proposed methods compared to Sobel, Prewitt and Roberts algorithms. Visual inspection of the results confirm that these algorithms produce the cleanest output as seen in Fig. 7.

V. DIFFICULTIES, FUTURE WORK AND CONCLUSION

A. Difficulties and Future Work

Throughout the completion of the project presented in this paper, difficulties were faced in its implementation and testing. These difficulties included determining the accuracy of a particular algorithm being tested. The challenge of comparing algorithms with various images is that an algorithm may be better suited to images with particular features. For this reason, visual inspection of the final output image is imperative to determine if the desired result is being obtained in conjunction with the analytical comparisons presented in this paper. The development of the fuzzy logic system was difficult due to the need for expert knowledge of fuzzy logic system design when developing these systems. Fuzzy logic systems need to be developed by a designer that knows how the system will behave, so extensive research was done to understand how to get accurate results from these systems.

Further development of the fuzzy rule base could possibly produce a better output image but further testing would need to be done to see if these improvements could be realized. Additional analysis of the accuracy of the algorithm could be conducted by researching more ways to quantify the algorithms effectiveness.

In addition to this, the thinning algorithm could possibly be improved by tuning its implementation to the photo being analysed. The implementation presented in this project uses a recursive thinning morphological operator provided by MATLAB. A custom tuned implementation could improve the output by not discarding edge data that should be in the output image.

Finally, the algorithm could be improved by performing the edge detection on the original coloured image without converting it to grayscale first. Doing so could improve the algorithms ability to detect edges of object with similar grayscale values on either side of the boundary. An example of this can be seen in Fig. 9b where the right side of the windows on the house are not being detected by the algorithm. Inspection of the original image shows that the grayscale values both inside and outside the window on the right side are very similar. This causes the fuzzy system to not indicate an edge. Performing the same edge detection algorithm on the colour image could fix this issue by allowing for the possibility of edges in the red, green and blue channels.

B. Conclusion

Through the successful implementation of this project, a fuzzy logic based edge detection algorithm was shown to produce similar results to Sobel, Prewitt, Roberts, Log, Zerocross and Canny based methods. Basing the implementation on a first-order derivation of the gradients in an image, this project has made improvements to the algorithm by showing that utilizing diagonal gradients produces a better result than an implementation using only horizontal and vertical gradients. In addition to visual inspection of results, comparisons were done to determine the effectiveness of the proposed solution against solutions presented in other sources. This was done through entropy calculations which gives insight into the apparent randomness of an image. Finally, it was shown that the proposed solution produces comparable results to other similar implementations of fuzzy logic based edge detection algorithms and that fuzzy logic is a viable solution to detect edges in an image. The completion of this project outlines and confirms the intuitive nature of fuzzy logic systems by presenting a design that

is easy to understand and produces the expected results.

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